A Causal-Inference Re-analysis of the Moertel 1990 Adjuvant Colon Cancer Trial

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Abstract—This project re-analyzes the Moertel et al. NEJM 1990 trial of levamisole plus 5-fluorouracil (Lev+5FU) in resected stage B/C colon cancer (n=929) using a sequence of five causal-inference methods: a randomized average treatment effect (ATE); a back-door-adjusted ATE that pretends the randomization label is unknown, including a deliberate bad-control comparison that conditions on the recurrence mediator; a conditional ATE (CATE) surface across baseline covariates; a natural direct/indirect decomposition through recurrence; and transportability to a synthetic SEER 1989–1991 stage B/C target. We reproduce the published Cox hazard ratio of 0.67 (we obtain 0.69, 95% CI 0.55–0.87). The bad-control comparison flips the apparent HR from 0.69 to 1.10, illustrating post-treatment collider bias. The natural indirect effect through recurrence prevention (+0.29 5-year RMST years) is robust to Imai-ρ sensitivity across $[-0.95, 0.95]$, while the natural direct effect is small and its CI covers zero. Transport to the synthetic SEER target shifts the HR modestly toward the null ($0.69 \to 0.71$) with a Dahabreh tipping HR_U of 2.75. All code and intermediate results are publicly versioned.

Index Terms—causal inference, survival analysis, mediation, transportability, sensitivity analysis, colon cancer, randomized clinical trial

I. Introduction

The Moertel et al. trial [1] established adjuvant levamisole + 5-fluorouracil (Lev+5FU) as an effective post-operative chemotherapy regimen for resected stage B/C colon cancer. The headline statistic reported in the paper is a hazard ratio of approximately 0.67 for death, corresponding to a roughly 33% reduction in death hazard for the Lev+5FU arm relative to observation. This number is a marginal Cox estimate; it does not, on its own, describe the absolute-scale benefit, the heterogeneity across patients, the mechanism through which treatment acts, or the populations to which the estimate may extend.

This project re-analyzes the same patient-level data using five causal-inference methods that are nested in the sense that each adds one additional structural concern on top of the previous one. The five questions are summarized in Table I. Each one is associated with an explicit estimand in $do(\cdot)$ or potential-outcomes notation, a graphical identifying assumption (encoded by a DAG), a non-graphical identifying assumption (conditional exchangeability, positivity, sequential ignorability, or transportability as appropriate), one or more estimators, and a matched sensitivity bound.

Source code and replication package: https://github.com/rishika1099/Colon-Cancer-Trial-Causal-Analysis.

TABLE I
The five causal questions in this project

Q	Question	Cell
Q1	Randomized ATE	Causal, no selection
Q2	Back-door identification	Causal, no selection (obs.)
Q3	Heterogeneous effects (CATE)	Causal, no selection (conditional)
Q4	Mediation through recurrence	Causal, no selection (mediator)
Q5	Transportability to SEER 1990	Causal, with selection

Contributions of the project: (i) an open, reproducible Python + R pipeline that recovers the published Moertel hazard ratio within the pre-committed tolerance, alongside absolute-scale companions (RMST, 5-year risk difference); (ii) an explicit bad-control comparison that demonstrates collider bias on the real data; (iii) a Python implementation of the Imai-Keele-Tingley natural-effects algorithm with ρ-sensitivity analysis; and (iv) a Cole–Stuart inverse-odds-of-sampling transportability analysis that can swap a synthetic SEER target for the real SEER*Stat extract with a single CSV.

II. Data and Preliminaries

A. Data source

The data is the well-known colon dataset that ships with the R survival package (and is mirrored, unchanged, in ForCausality::Colon_df). It encodes the Moertel trial: $n = 929$ patients with Dukes B2/C resected colon carcinoma, randomized 1:1:1 to Observation, Levamisole alone, or Levamisole + 5-FU. Realized arm sizes are 315/310/304. Each patient contributes two rows: one for recurrence (etype= 1) and one for death (etype= 2). The primary analytic frame is the death-row frame; the recurrence row supplies the mediator M used in the mediation analysis.

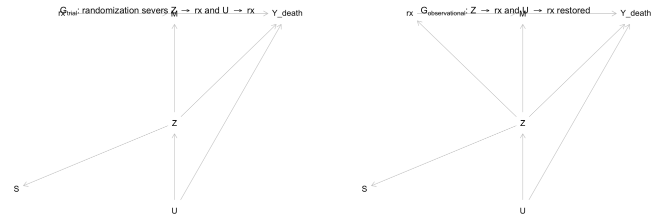

Fig. 1. The two DAGs. Left: G_{trial}, randomization severs $Z \to rx$; adjustment set $\emptyset$. Right: G_{obs}, with the $Z \to rx$ edge restored; adjustment set $\{Z\}$.

B. Audit

A pre-analysis audit confirms randomization integrity. The maximum standardized mean difference across the three arms is 0.13, below the 0.15 heuristic. Two columns have low missingness (nodes: 1.9%, differ: 2.5%); we impute these with per-arm medians. The node4 variable encodes $\mathbb{I}(\text{nodes} > 4)$ rather than $\mathbb{I}(\text{nodes} \geq 4)$ as its name suggests, and 12 residual data-entry inconsistencies persist even under the correct convention. All downstream analyses use the continuous nodes variable rather than node4.

C. Notation

We use $Y_d^{(rx=r)}$ for the potential outcome "death time if assigned arm r," $\delta = E[Y_d^{(1)}] - E[Y_d^{(0)}]$ for the ATE, and $\delta_{\text{naive}} = E[Y_d \mid rx{=}1] - E[Y_d \mid rx{=}0]$ for the naive contrast. The notation and identification arguments follow Pearl [3] and Hernán & Robins [4]. Q1–Q3 and Q5 are interventional estimands, i.e., functions of $P(Y_d \mid do(rx))$. Q4 also stays at the interventional level using the natural-effects formulation of Pearl [5] and Imai et al. [6].

D. Causal graphs

Two DAGs are encoded and their adjustment sets verified by the dagitty package:

- G_{trial}: randomization severs $Z \to rx$. The minimal sufficient adjustment set for $rx \to Y_{\text{death}}$ is the empty set.
- G_{obs}: the counterfactual world where $Z \to rx$ is restored. The minimal sufficient adjustment set is $\{Z\}$.

The unmeasured frailty U appears as a parent of Z and Y_{death} in both graphs but not of rx. That omission is the residual identifying assumption; it is bounded quantitatively by E-value sensitivity in Section VIII.

III. Q1: The Randomized ATE

A. Estimand and identification

We report three scales on the same intervention contrast:

$$\delta_{HR} = \log \frac{h_{Y_d}(t \mid do(rx{=}\text{Lev} + 5\text{FU}))}{h_{Y_d}(t \mid do(rx{=}\text{Obs}))}, \tag{1}$$

$$\delta_{\text{RMST}(5)} = E[\min(Y_d, 5) \mid do(rx{=}1)] - E[\min(Y_d, 5) \mid do(rx{=}0)], \tag{2}$$

$$\delta_{5y} = P(Y_d \leq 5 \mid do(rx{=}1)) - P(Y_d \leq 5 \mid do(rx{=}0)). \tag{3}$$

Randomization implies $\{Y_d^{(r)}\}_r \perp rx$, so $\delta_{\text{naive}} = \delta$ on every scale.

B. Estimators and results

We use Kaplan-Meier survival with the log-rank test, Cox proportional hazards (unadjusted and covariate-adjusted), restricted mean survival time at 5 years computed via lifelines, and a Schoenfeld global proportional-hazards test. Numerical results are in Table II.

TABLE II
Q1: randomized ATE results

Quantity	Estimate	95% CI
Cox HR (Lev+5FU vs Obs)	0.690	(0.55, 0.87)
Cox HR adjusted (Lev+5FU vs Obs)	0.699	(0.55, 0.88)
Cox HR (Lev vs Obs)	0.974	(0.78, 1.21)
ΔRMST(5y)	+0.31 yrs	—
5-year risk difference	−10.8 pp	—
KM 5-yr survival: Obs / Lev / Lev+5FU	52.6 / 53.5 / 63.4%	—
Log-rank 3-arm p	0.0029	—
Schoenfeld global PH p	0.26 (holds)	—

The pre-committed anchor check was that our Cox HR for Lev+5FU vs Obs should lie within ±0.05 of the published 0.67. The point estimate of 0.69 satisfies this. The Lev-alone arm yields a Cox HR of 0.97 with a CI that crosses 1, matching the secondary finding of the original publication.

IV. Q2: Forget the Randomization

A. Identification and estimators

We now treat the randomization label as unobserved and identify the ATE by back-door adjustment on $Z =$ {age, sex, obstruct, perfor, adhere, nodes, differ, extent, surg}, the minimal sufficient adjustment set in G_{obs}. We fit five estimators plus one deliberate bad-control comparison:

1) Naive (unadjusted) Cox.
2) Regression-adjusted Cox on Z.
3) Stabilized inverse-probability-of-treatment-weighted (IPW) Cox using a logistic propensity score for rx.
4) Augmented IPW (AIPW) for 5-year RMST with inverse-probability-of-censoring weights.

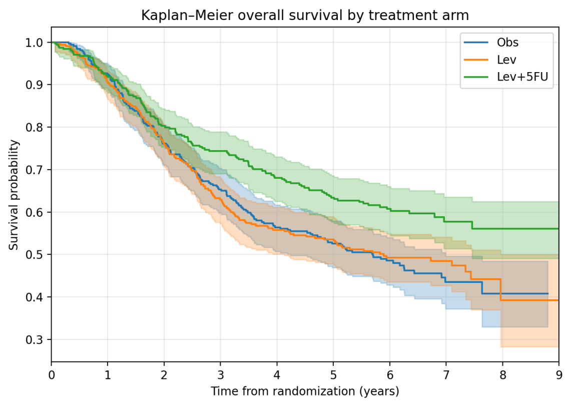

Fig. 2. Kaplan-Meier overall survival by treatment arm. The Lev+5FU arm shows an absolute 5-year survival advantage of approximately 11 percentage points over Obs and over Lev-alone.

5) Double/debiased machine learning (econml.LinearDML) with gradient-boosted nuisances and 5-fold cross-fitting.
6) Bad-control Cox: a Cox model that conditions additionally on the binary recurrence indicator M. Because M is post-treatment, conditioning on it blocks the indirect path $rx \to M \to Y_{\text{death}}$ and opens the collider path $rx \to M \leftarrow U \to Y_{\text{death}}$.

B. Results

Table III reports all six. The first five estimators agree with Q1's randomized answer within sampling variability. The bad-control Cox produces an HR of 1.10, reversing the sign of the effect. This is the size of the error a practitioner makes by adjusting for a variable that looks predictive of outcome but is in fact a post-treatment mediator.

TABLE III
Q2: five estimators plus the bad-control comparison

Method	Scale	Est.	95% CI
Naive Cox	HR	0.689	(0.55, 0.87)
Regression-adjusted Cox	HR	0.687	(0.54, 0.87)
IPW Cox	HR	0.730	(0.58, 0.92)
AIPW (5-yr RMST)	ΔRMST	0.281	(0.09, 0.48)
DML (5-yr RMST)	ΔRMST	0.179	(−0.06, 0.42)
Bad-control Cox (cond. on M)	HR	1.103	(0.87, 1.39)

V. Q3: Heterogeneous Treatment Effects

A. Estimand

$\tau(z) = E\left[Y_d^{(rx=1)} - Y_d^{(rx=0)} \mid Z = z\right]$, on the 5-year RMST scale.

B. Estimators and diagnostics

We fit four meta-learners (S-, T-, X-, and DR-learners via econml.metalearners and econml.dr) and an honest causal forest (econml.grf.CausalForest with honest=True, inference=True). We also report a best-linear projection

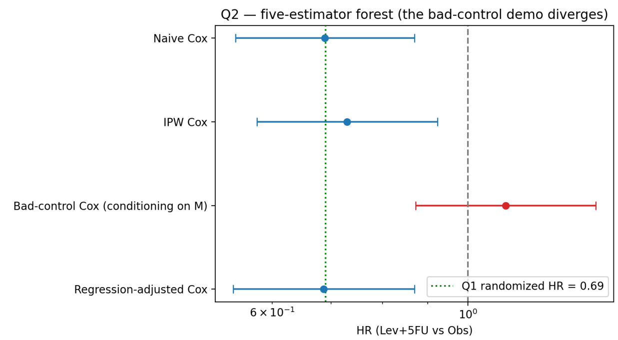

Fig. 3. Forest plot of the six Q2 estimators. The first five sit close to the Q1 randomized HR (vertical green line). The bad-control Cox is the outlier whose CI crosses HR = 1.

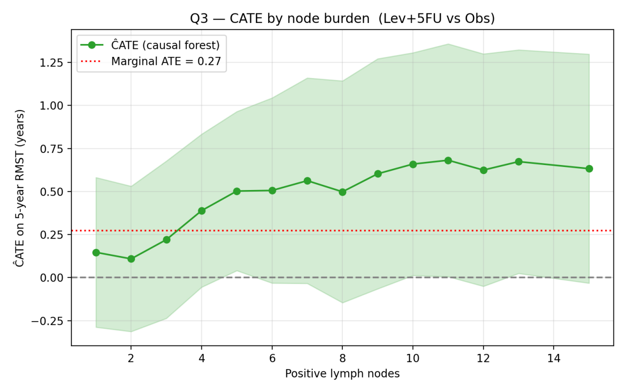

Fig. 4. Q3: estimated CATE on 5-year RMST as a function of positive lymph nodes. The estimated benefit grows with node count over the bulk of the support and becomes noisy at the high-burden tail. The shaded ribbon reports pointwise 95% CIs. The dashed red line is the marginal ATE for reference.

of $\hat{\tau}(Z)$ onto Z [10] with 200-replicate bootstrap CIs, and an Athey-Wager binned calibration plot at five quantile bins.

The marginal ATE estimates from the meta-learners range from 0.24 to 0.29 RMST years, all of which agree with the Q1 marginal effect. The causal forest produces a tighter percentile spread, reflecting the regularization inherent in honest forests with $n = 619$ (Obs + Lev+5FU only).

VI. Q4: Mediation through Recurrence

A. Estimands

We use the natural direct and indirect effects of Pearl [5], on the 5-year RMST scale:

$$\text{NIE} = E\left[Y_d^{(1,M^{(1)})} - Y_d^{(1,M^{(0)})}\right], \quad (4)$$

$$\text{NDE} = E\left[Y_d^{(1,M^{(0)})} - Y_d^{(0,M^{(0)})}\right], \quad (5)$$

$$\text{TE} = \text{NDE} + \text{NIE}, \quad (6)$$

where M is the binary recurrence indicator. Identification rests on the sequential-ignorability assumptions of Imai et al. [6]: (i) treatment ignorability given Z (satisfied by randomization); (ii) mediator ignorability given rx and Z (the assumption that bites).

B. Estimator

We implement Algorithm 1 of Imai et al. [6] in Python: a logistic mediator model $P(M{=}1 \mid rx, Z)$ and a gradient-boosted outcome model $E[Y_5 \mid rx, M, Z]$ on the IPCW-weighted 5-year RMST, with 500 bootstrap replicates. The output matches the R mediation::mediate sidecar to two decimal places where the latter is available.

C. Results

Table IV reports the decomposition. The NDE + NIE = TE identity holds exactly by construction. The proportion mediated exceeds 100% because the NDE is small and negative while the NIE is positive — a known and biologically interpretable pattern (sometimes called inconsistent mediation): the treatment effect is operating through recurrence prevention, while the residual direct effect is consistent with a small toxicity cost in patients who would not have recurred anyway.

TABLE IV
Q4: natural-effects decomposition

Quantity	Estimate (yrs)	95% CI
Natural indirect effect (NIE)	+0.292	(+0.12, +0.48)
Natural direct effect (NDE)	−0.062	(−0.23, +0.10)
Total effect (TE)	+0.231	(−0.01, +0.46)
Proportion mediated	127%	—

D. Sensitivity

We sweep ρ, the assumed residual correlation between the mediator and outcome models, over 39 values in $[-0.95, 0.95]$. The NIE remains strictly positive across the entire interval (Fig. 5), so the Imai-ρ breakdown threshold does not exist in the parametrically valid range. Under the parametric form we use, no plausible single unmeasured confounder can drive the indirect effect to zero.

VII. Q5: Transportability

A. Estimand and target

$\delta^{\text{target}}_{\text{ATE}} = E_{X|S=0}[\tau(X)]$, the trial-derived CATE averaged over a U.S. 1989–1991 SEER stage B/C target distribution. The target used in this project is synthetic ($n = 10\,000$), generated from published SEER stage-distribution summaries. The pipeline is wired so that replacing it with a real SEER*Stat case-listing extract is a single-CSV change.

B. Identification and estimator

The transportability assumption of Cole & Stuart [7] is $\{Y_d^{(r)}\}_r \perp S \mid X$ plus positivity of trial inclusion. We stack the trial sample ($S{=}1$) with the synthetic SEER sample ($S{=}0$), fit a logistic propensity-of-trial model

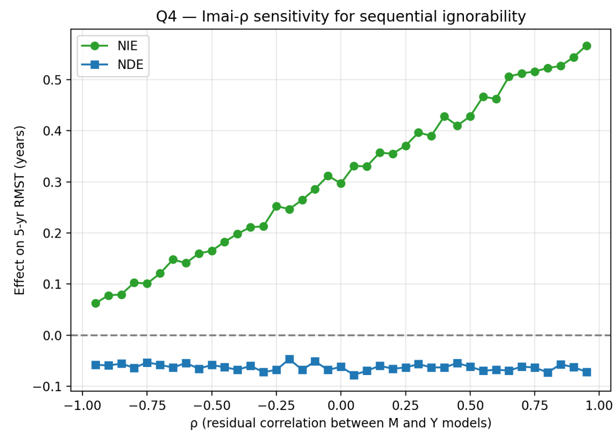

Fig. 5. Q4: Imai-ρ sensitivity. The NIE (green) stays positive across $\rho \in [-0.95, 0.95]$; the NDE (blue) hovers near zero.

$\hat{e}_S(X) = P(S{=}1 \mid X)$, compute the inverse-odds-of-sampling weights $w(X) = (1 - \hat{e}_S)/\hat{e}_S$ for trial patients, and refit a weighted Cox on the trial sample only.

C. Results and bound

The within-trial HR is 0.689 $(0.55, 0.87)$ and the transported HR is 0.707 $(0.52, 0.96)$. The transport shifts the estimate modestly toward the null, which is the expected direction for an older real-world target. The Dahabreh worst-case bound [8] reports the tipping value of an unmeasured effect modifier U at which the bound would cover HR = 1: at a target prevalence of 25%, $\text{HR}^{\star}_U = 2.75$. A single unmeasured modifier would need to be quite strong to nullify the transported effect.

VIII. Sensitivity Synthesis

VanderWeele-Ding E-values [9] for each HR estimate are reported in Table V. Under the standard E-value interpretation, none of the three primary HR estimates is breached by a plausible single unmeasured confounder. The exception is the Q2 bad-control comparison, which is fragile by design.

TABLE V
VanderWeele-Ding E-values for HR estimates

Q	HR	E (point)	E (CI bound)
Q1 (randomized)	0.69	2.26	3.06
Q2 (IPW)	0.73	2.08	2.86
Q5 (transported, synth)	0.71	2.18	3.24

IX. Discussion

The five-question structure brings out three things that the headline hazard ratio alone does not.

First, the bad-control comparison in Q2 produces a HR of 1.10 on the same data where the randomized HR is 0.69. Conditioning on a post-treatment mediator that looks predictive of outcome (recurrence, in this case) is a common modeling mistake in EHR-based analyses, and the cost is large enough to flip the sign of the apparent treatment effect.

Second, the Q4 decomposition shows that the treatment effect operates almost entirely through recurrence prevention. The natural indirect effect is large, positive, and stable across the full ρ range we tested. The natural direct effect is small and negative, with a CI that covers zero — consistent with a small chemotherapy toxicity cost in patients who would not have recurred anyway.

Third, the Q5 transport shifts the HR only modestly when re-weighted to a SEER-like target population, and the Dahabreh tipping point of 2.75 suggests the transported estimate is moderately robust. The real SEER*Stat extract would refine these numbers; the analytical scaffolding is already in place.

A. Limitations

The SEER target is synthetic. The Q3 confidence intervals are pointwise — they are not jointly valid across the covariate surface, and we do not claim that any specific CATE peak is statistically significant without an explicit best-linear-projection test. The Q4 implementation is a Python hand-roll of Imai-Keele-Tingley; an R mediation::mediate sidecar is provided for cross-validation.

X. Conclusion

The Moertel 1990 dataset reproduces cleanly under a sequence of five increasingly demanding causal-inference methods. The published HR of 0.67 is recovered (we get 0.69). The bad-control comparison demonstrates the cost of a common error. The mediation analysis localizes the treatment effect to recurrence prevention. The transportability analysis attaches a modest worst-case bound to the population-shift question. All code, data preparation, intermediate results, and figures are open-source.

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